

REPORT
ON
EIGHT WEEKS OF INTERNSHIP – II
(Research / Industry Internship)

Carried out at
**Nitte Centre of Excellence for Applied Artificial
Intelligence**

Submitted by
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To



(Estd. under Section 3, UGC Act 1956)
(Placed under Category 'A' by MHRD, Govt. of India; Accredited as 'A+' Grade University by NAAC)

In partial fulfilment of the requirements for the award of
**BACHELOR OF TECHNOLOGY in INFORMATION SCIENCE
AND ENGINEERING**

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING



June 2025–August 2025



NITTE
(Deemed to be University)

**NMAM INSTITUTE
OF TECHNOLOGY**

Department of Information Science and Engineering

CERTIFICATE

This is to certify that **Internship-II (Research Internship / Industry Internship)** report is a bonafide record of the work carried out by **Ms Pratheeksha Nayak (NNM22IS116)**. She has satisfactorily completed eight weeks of internship during **June 2025 to August 2025** as part of the B.Tech regulations. This work fulfills the partial requirements for the award of the degree of **Bachelor of Technology in Information Science and Engineering** from Nitte Deemed to be University, Mangaluru

Name and Signature of Mentor

Signature of HOD

Semester End Viva Voce Examination

Name of the Examiners

Signature with Date

1. _____

2. _____

Training Certificate:



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I would like to express my sincere gratitude to the **Nitte Center of Excellence for Applied Artificial Intelligence (CoE-AAI)** at **NMAM Institute of Technology** for providing me the opportunity to undertake my internship. Working on the project “Fine-Tuning Transformer Models for Zero-Shot Image Classification in Low-Resource Medical Image Diagnosis” has been an invaluable experience that allowed me to bridge the gap between theoretical Fine-Tuning Transformer Models concepts and practical, real-world application.

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Abstract

The internship was carried out at the Nitte Centre of Excellence for Applied Artificial Intelligence, a research and development center dedicated to advancing artificial intelligence and machine learning technologies through applied research and innovation. The center focuses on bridging theoretical computer science with practical, industry-ready intelligent systems and contributes significantly to emerging developments in healthcare, automation, and intelligent data-driven applications. Its competitive advantage lies in integrating academic research with real-world problem-solving methodologies, enabling students and researchers to work on contemporary AI applications under expert mentorship and advanced computing infrastructure. The primary objective of the internship was to develop a system for brain MRI tumor classification using transformer-based vision–language models under low-resource conditions, while exploring the effectiveness of zero-shot learning, prompt engineering, and few-shot fine-tuning techniques for classifying MRI images into four tumor categories. Initially, a pretrained vision–language model was applied for zero-shot classification by comparing image features with textual descriptions of tumor classes. Domain-specific prompts and a partial transformer fine-tuning strategy were then introduced using a small labeled dataset, demonstrating that while zero-shot inference yields limited accuracy due to domain differences, even minimal fine-tuning significantly improves classification performance. During the internship, innovative practices such as prompt engineering, selective transformer adaptation, and lightweight data augmentation techniques were implemented to improve model adaptability under constrained data conditions. A key area identified for future improvement is the optimization of models for faster execution, enhanced computational efficiency, and improved generalization on larger medical datasets. Ultimately, this internship bridged the gap between theoretical knowledge and practical software engineering, strengthened understanding of ethical data handling, professional conduct, collaborative research practices, and significantly enhanced technical competency, research aptitude, and future employment readiness while successfully fulfilling the core objectives of the B.Tech program.

1. Introduction of the Research Centre

1.1 History and Overview

The Nitte Centre of Excellence for Applied Artificial Intelligence (CoE-AAI) is an innovative research initiative established within the NMAM Institute of Technology (NMAMIT), Nitte. It was founded to serve as a specialized hub for exploring the practical applications of Artificial Intelligence and Machine Learning in both academic and industrial domains. The centre functions as a dedicated research and development wing, bridging the gap between theoretical computer science and real-world implementation by focusing on intelligent systems and advanced AI-driven solutions. By integrating modern technologies such as machine learning, natural language processing, and data analytics, the centre addresses practical challenges in automation and assessment quality.

1.2 Nature of Center and Size

Operating within the academic ecosystem of Nitte (Deemed to be University), the centre's primary business revolves around applied research, academic automation, and intellectual property development. It is a mid-sized, specialized research laboratory that facilitates collaboration between the Department of Information Science and Engineering and external industry partners. The organization provides a structured environment where faculty mentors, research scholars, and student interns work together to develop educational technologies that enhance curriculum alignment and assessment reliability.

1.3 Organizational Chart

The administrative and technical structure of the CoE-AAI is designed to ensure research integrity and systematic project execution:

- Principal, NMAMIT: Provides institutional leadership and final project approvals.
- Director / Head of CoE-AAI: Manages the strategic direction, resource allocation, and overall research agenda.
- Faculty Mentors / Subject Matter Experts: Provide specific technical guidance and supervise research tracks (Dr. Roopa B Hegde and Dr. Srinivasa Pai P.).

- **Research Interns:** Responsible for the hands-on design, development, and testing of specific software modules and prototypes.

1.4 Types of Products Produced and Services Offered

The CoE-AAI operates as a collaborative research environment rather than a traditional manufacturing or software product company. Its primary ‘products’ are research prototypes, intelligent software systems, and predictive models. The services offered include applied research, model training, dataset analysis, and the development of AI-driven solutions across a wide array of disciplines. The centre produces research outputs across several key domains:

- **Healthcare and Medical AI:** Development of diagnostic and predictive tools, including CVM-based skeletal age prediction using gender-aware ML models, multimodal explainable AI for healthcare decision support, and AI systems for optimizing chest CT scan bolus tracking arrival times. It also develops secure applications like RADAccess for fingerprint-based visualization of cloud-hosted radiology reports.
- **Natural Language Processing (NLP):** Designing text-analysis systems, ranging from the Question Paper Generator using NLP to models for Restaurant Review Analysis.
- **Materials Science and Engineering:** Utilizing machine learning for advanced materials discovery and property prediction specifically tailored for aerospace and marine applications.
- **Agriculture and Environment:** Developing machine learning-driven solutions for crop disease detection using hyperspectral remote sensing, and creating virtual AI-based immersive herbal gardens.
- **Predictive Analytics:** Building forecasting models, such as machine learning systems for predicting Ola bike ride requests.

1.5 Competitive Advantage and Core Competency

The core competency of the centre lies in its versatile ability to apply diverse artificial intelligence and machine learning techniques spanning Natural Language Processing (NLP), computer vision,

and predictive analytics to solve complex, real-world problems across multiple domains such as healthcare, agriculture, and materials science. A major competitive advantage is the centre's recent investment in high-performance computing infrastructure, specifically the addition of three new high-computing desktops equipped with advanced multi-core processors and high-end dedicated graphics cards, which significantly accelerate the training of complex, resource-intensive machine learning models. This robust infrastructure allows the research teams to efficiently process massive, multimodal datasets, such as hyperspectral remote sensing data and high-resolution medical images, providing a superior environment for developing innovative solutions ranging from text-based automation to highly complex applications like fine-tuning transformer models for medical image diagnosis and materials property prediction.

2. Details of the Training Completed

2.1 Overview of the Project

During the eight-week internship at the Nitte Centre of Excellence for Applied Artificial Intelligence (CoE-AAI), NMAM Institute of Technology, work was carried out on a research project titled “Fine-Tuning Transformer Models for Zero-Shot Image Classification in Low-Resource Medical Image Diagnosis.” The project focused on applying modern deep learning techniques for medical image analysis, particularly for identifying different types of brain tumors from MRI scans. In medical imaging applications, obtaining large labeled datasets is difficult because annotation requires expert radiologists and strict privacy regulations. Therefore, the main objective of this project was to investigate whether pretrained vision–language transformer models could classify medical images effectively even when only a small number of labeled samples are available. The project explored different learning approaches including zero-shot learning, prompt engineering, and few-shot fine-tuning, aiming to develop a data-efficient classification framework capable of assisting medical professionals in analyzing brain MRI scans.

2.2 Medical Image Dataset Preparation

The dataset used in this project consisted of brain MRI images categorized into four diagnostic classes: glioma tumor, meningioma tumor, pituitary tumor, and normal brain images with no tumor. These images represent common brain tumor conditions observed in clinical neuroimaging. MRI scans contain complex grayscale patterns and subtle intensity variations that make automated classification challenging. Before training the model, several preprocessing steps were applied: the MRI images were resized to a fixed resolution and converted into three-channel RGB format to ensure compatibility with the pretrained model architecture. Standard normalization techniques were also applied to align the images with the preprocessing pipeline used during the pretraining of the CLIP vision encoder.

2.3 Zero-Shot Classification Module

The first stage of the project involved implementing a zero-shot classification pipeline using a pretrained vision–language transformer model. In this approach, the model performs classification without using any labeled medical training data, relying instead on knowledge learned during large-scale pretraining on image-text pairs. Each MRI image was passed through the vision encoder to generate a feature representation. At the same time, textual descriptions corresponding to each tumor class were processed using the text encoder to generate semantic embeddings. The similarity between the image representation and each class description was computed using cosine similarity, and the class with the highest similarity score was selected as the predicted label. Although this method demonstrates the generalization ability of vision–language models, the performance was limited due to the domain difference between natural images and medical MRI scans.

2.4 Prompt Engineering Module

To improve the performance of zero-shot classification, a prompt engineering strategy was introduced. Instead of using simple class labels such as “glioma tumor” or “pituitary tumor,” multiple domain-specific textual prompts were created for each class, including medically relevant descriptions related to tumor appearance and MRI imaging context. Each prompt was encoded separately using the text encoder, and the resulting embeddings were averaged to form a robust representation of each tumor class. This multi-prompt approach helped reduce the sensitivity of the model to individual prompt wording and improved the alignment between visual features and textual descriptions. Although prompt engineering improved classification performance slightly, it was still insufficient to fully overcome the domain mismatch between natural images and medical MRI scans.

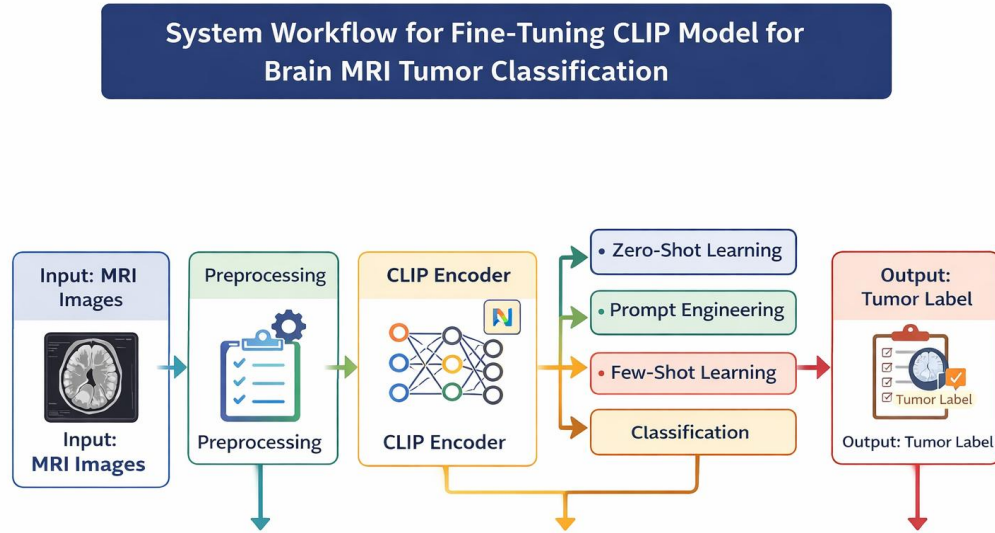


Fig. 1. Overview of the proposed framework illustrating zero-shot inference, prompt engineering, few-shot learning, and partial transformer adaptation.

2.5 Few-Shot Learning and Model Adaptation

To further improve classification accuracy, the project implemented a few-shot learning strategy using only 20 labeled MRI images per class. Initially, the pretrained vision encoder was kept frozen and a simple linear classifier was trained on top of the extracted image features. However, the results showed that the frozen backbone approach produced only marginal improvement. Therefore, a partial transformer adaptation strategy was introduced, where only the final transformer block of the vision encoder was unfrozen and trained together with the classification layer. This selective fine-tuning allowed the model to adapt to the specific characteristics of medical images while preserving the general knowledge learned during pretraining.

2.6 Implementation and Model Training

The system was implemented using the PyTorch deep learning framework, and experiments were conducted using GPU acceleration on the Google Colab platform. The backbone architecture used in the project was the CLIP model with a Vision Transformer (ViT-B/32) encoder. To reduce overfitting caused by the limited dataset, lightweight data augmentation techniques were applied during training, including random cropping, horizontal flipping, slight rotations, and brightness adjustments. The AdamW optimizer was used for model optimization, and training was performed for a limited number of epochs to prevent overfitting.

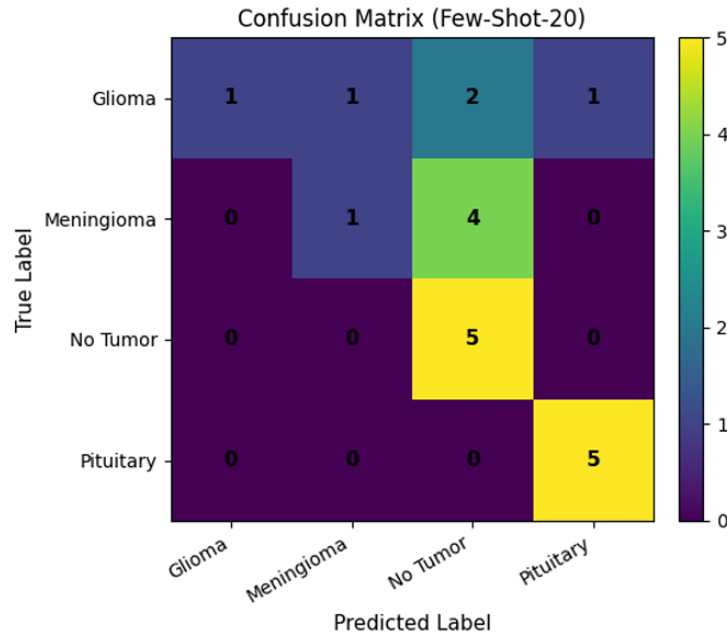
2.7 Performance Evaluation and Results

The performance of the proposed system was evaluated using several classification metrics, including accuracy, precision, recall, and F1-score. The baseline zero-shot model achieved approximately 28% accuracy, which is close to random performance for a four-class classification task. Introducing domain-specific prompt engineering improved the accuracy to around 38%, indicating that improved textual descriptions can enhance semantic alignment between image and text representations. However, the most significant improvement was observed when partial transformer fine-tuning was applied: with only twenty labeled images per class, the partially fine-tuned model achieved approximately 60% classification accuracy, demonstrating the effectiveness of minimal model adaptation for medical image classification.

TABLE I: Performance comparison across different learning settings

Method	Training Data	Accuracy
Zero-shot CLIP	0	28%
Prompt-engineered CLIP	0	38%
Frozen Few-shot CLIP	20/class	30%
Partial Transformer Adaptation	20/class	60%

The confusion matrix shows that the model achieved strong performance for the pituitary tumor and normal brain classes, while some confusion occurred between glioma and meningioma tumors due to their similar MRI characteristics.

*Fig. 2. Confusion matrix showing class-wise performance of the proposed model.*

2.8 Overall Learning Outcome

During the internship, practical experience was gained in several important areas of artificial intelligence and machine learning:

- Understanding zero-shot and few-shot learning techniques
- Working with vision–language transformer models
- Implementing deep learning pipelines using PyTorch
- Preprocessing and analyzing medical imaging datasets
- Applying prompt engineering strategies
- Evaluating machine learning models using statistical performance metrics

In addition to technical skills, the internship improved the ability to approach research problems systematically and to design experiments that validate machine learning models under real-world constraints.

2.9 Future Works and System Extensibility

Although the proposed approach achieved promising results, several improvements can be explored in future work:

1. The system can be evaluated on larger and more diverse medical imaging datasets to improve generalization across different clinical conditions. Additionally, integrating multimodal clinical data such as radiology reports or patient history may further improve diagnostic performance.
2. Another potential direction is the use of parameter-efficient fine-tuning techniques, such as adapter modules or LoRA-based tuning, which can further reduce computational requirements while maintaining strong performance.
3. Finally, integrating visual explainability techniques, such as attention maps or Grad-CAM visualizations, could help clinicians better understand the model’s decision-making process

3. Conclusion

The internship at the Nitte Centre of Excellence for Applied Artificial Intelligence provided valuable exposure to the practical application of modern deep learning techniques in medical image diagnosis through the development of a transformer-based vision–language model for brain MRI tumor classification under zero-shot and low-resource learning conditions. The project enabled practical understanding of how advanced artificial intelligence models can be adapted for real-world healthcare applications where labeled medical data is limited. The experimental study revealed that while direct zero-shot classification is insufficient for medical imaging tasks due to domain mismatch, incorporating domain-specific prompt engineering and minimal partial transformer fine-tuning with as few as twenty labeled images per class significantly improved classification accuracy from 28% to approximately 60%. During the internship, exposure was gained in dataset preprocessing, model training, performance evaluation, and experimental research methodology using PyTorch and transformer-based architectures. The internship also strengthened problem-solving ability, analytical thinking, and understanding of the practical implementation of theoretical machine learning concepts under real-world constraints. In addition to technical knowledge, the training emphasized the importance of professional ethics, collaborative research, effective communication, and responsible handling of medical data in AI-driven healthcare systems. Although the proposed system achieved promising results, further improvements such as optimizing model execution speed, integrating multimodal clinical information, and incorporating explainable AI techniques can enhance future system performance and clinical reliability. Overall, the internship significantly enhanced technical competency, research aptitude, and professional readiness, while successfully fulfilling the core objectives of the B.Tech program and improving preparedness for future employment opportunities in the fields of artificial intelligence and data science.

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